සියලුම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடை யது/ All Rights Reserved]

Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmaw PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSICS Sanjeewa Dharmawardhana PHYSIC Sanieewa Dharmawardhana PHYSICS Sanieewa Dharmawardhana PHYSICS Sanieewa Dharmawardhana PHYSICS Sanieewa

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2027 අගෝස්තු
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2027 ஓகஸ்ட்
General Certificate of Education (Adv. Level) Examination, August 2027

භෞතික විද්‍යාව I
பௌதிகவியல் I
Physics I

Unit 02-Kinematics
Past Paper -01

01 E I

Name: *Institute.:*

Instructions:

* *This question paper consists of **50** questions in **12** pages.*
* *Answer all the questions*
* *Write your **Institute** in the space provided in the answer sheet.*
* *Read the instructions given on the back of the answer sheet carefully.*
* *In each of the questions **1** to **50**, pick one of the alternatives from* (1), (2), (3), (4), (5) *which is correct or **most appropriate** and **mark your response on the answer sheet with a cross (X) in accordance with the instructions given on the back of the answer sheet**.*

Use of calculators is not allowed

(Acceleration due to gravity, $g = 10\ \mathrm{N\ kg^{-1}}$) Duration :- **2 hrs**

1. A car and a bus stop at a red-light signal. The car is 100 m behind the bus. When the light turns green, the car accelerates with $6\ \mathrm{m\ s^{-2}}$ and at the same time the bus accelerates with $4\ \mathrm{m\ s^{-2}}$. Car will overtake the bus after,
   (1) $4\ s$ (2) $6\ s$ (3) $8\ s$ (4) $10\ s$ (5) 12 s

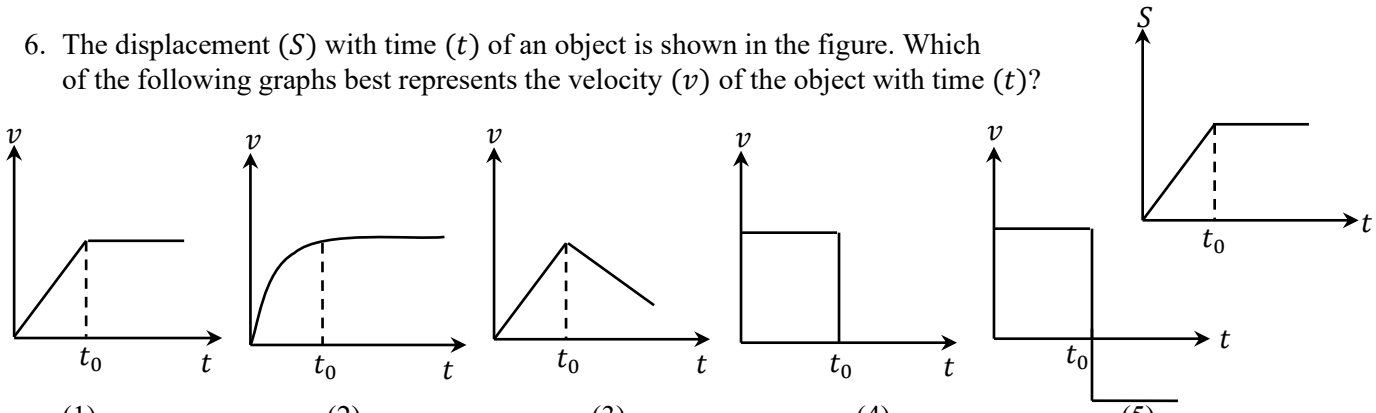
2. When two objects A and B move with uniform speeds toward each other along a straight line, they get 5 m closer to each other every second. If they move in the same direction along a straight line with the original speeds, they get 1 m closer to each other every second. The speeds of A and B are respectively
   (1) $5\ \mathrm{m\ s^{-1}}$ and $4\ \mathrm{m\ s^{-1}}$ (2) $5\ \mathrm{m\ s^{-1}}$ and $10\ \mathrm{m\ s^{-1}}$ (3) $3\ \mathrm{m\ s^{-1}}$ and $2\ \mathrm{m\ s^{-1}}$
   (4) $3\ \mathrm{m\ s^{-1}}$ and $1\ \mathrm{m\ s^{-1}}$ (5) $2\ \mathrm{m\ s^{-1}}$ and $1\ \mathrm{m\ s^{-1}}$

3. A particle P moving with a uniform velocity of $4\ \mathrm{m\ s^{-1}}$ along $x-$ axis passes the origin O at time $t = 0$.A second particle Q moving along the same direction with a uniform velocity of $5\ \mathrm{m\ s^{-1}}$ passes origin O at $t = 1$ s. Particle Q will reach the particle P when they have travelled a distance of
   (1) 10 m from the origin. (2) 16 m from the origin. (3) 20 m from the origin.
   (4) 25 m from the origin. (5) m from the origin.

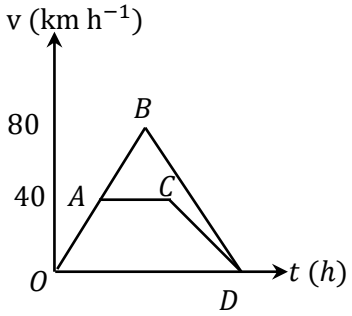
4. A particle moving at a constant velocity v passes a certain point at $t = 0$. If another particle starting from rest at $t = 0$ from the same point move towards the first particle with constant acceleration a, how long will the second particle take to reach the first particle?
   (1) $\frac{a}{v}$ (2) $\frac{a^2}{2v^2}$ (3) $\frac{2v}{a}$ (4) $\frac{2v^2}{a^2}$ (5) $\frac{v^2}{a^2}$

5. Two athletes run a 10 km race with constant speeds v_1 and v_2 in a circular track of radius 50 m. It has been observed that the athlete with speed v_1 completed 10 rounds when the other athlete completed 9 rounds.
   The ratio $\frac{v_1}{v_2}$ is
   (1) $\frac{10}{9}$ (2) $\frac{9\pi}{10}$ (3) $\frac{18\pi}{10}$ (4) $\frac{10\pi}{9}$ (5) 9

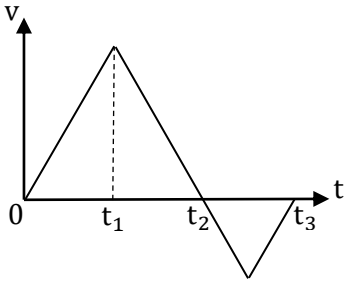
6. The displacement (S) with time (t) of an object is shown in the figure. Which of the following graphs best represents the velocity (v) of the object with time (t)?



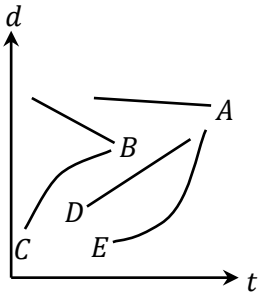
7. The speed (v) – time (t) curves of two similar private buses P and Q overloaded with passengers are shown in the figure. $OABD$ represents the $v - t$ curve for the bus P and $OACD$ represents the $v - t$ curve for the bus Q. Consider the following statements.
   (A) Retardation of bus P is greater than that of bus Q.
   (B) Both buses have travelled equal distances during the journey.
   (C) Standing passengers in bus Q feel more uncomfortable than those in P during the journey.
   Of the above statements
   (1) only (A) is true.
   (2) only (B) is true.
   (3) only (A) and (B) are true.
   (4) only (B) and (C) are true.
   (5) all (A), (B) and (C) are true.



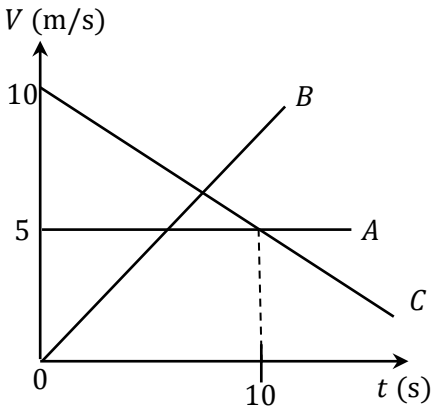
8. The figure shows the velocity – time graph for a particle which starts from rest and moves along $x -$ direction. According to this graph,
   (A) the particle comes to rest only at time $t = t_3$.
   (B) the particle returns to its original position at time $t = t_3$.
   (C) the particle accelerates only during the time interval $0 - t_1$
   Of the above statements
   (1) only (A) is true. (2) only (B) is true. (3) only (A) and (C) are true.
   (4) only (A) and (B) are true. (5) all (A), (B) and (C) are false.



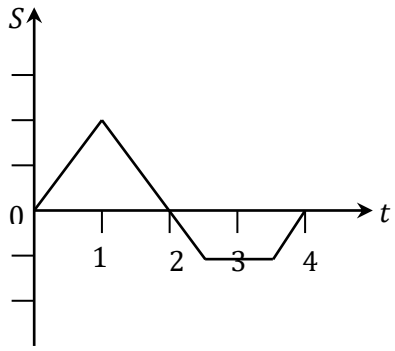
9. Figure shows five displacement (d) – time (t) curves for five different objects. The object which has an acceleration in direction of its motion is represented by
   (1) A (2) B (3) C
   (4) D (5) E

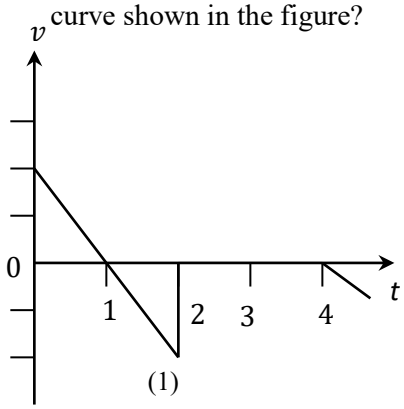


10. Figure shows velocity (V) – time (t) curves of three particles A, B and C moving along a straight line. If at $t = 0$ all the particles can be found together at a certain point on the straight line, then at $t = 10$ s
   (1) particles A and B meet again.
   (2) particles B and C meet again.
   (3) particles C and A meet again.
   (4) particles A, B and C all meet again.
   (5) none of the particles meet again.

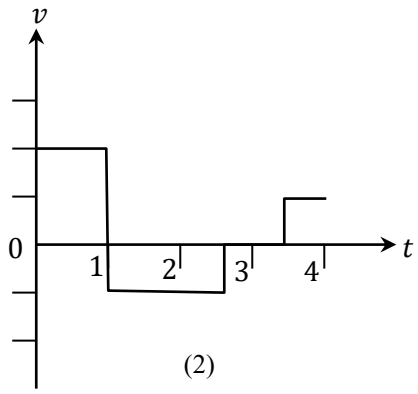


11. Which of the following curves correctly represents the corresponding velocity (v) – time (t) curve for the displacement (s) – time (t) curve shown in the figure?

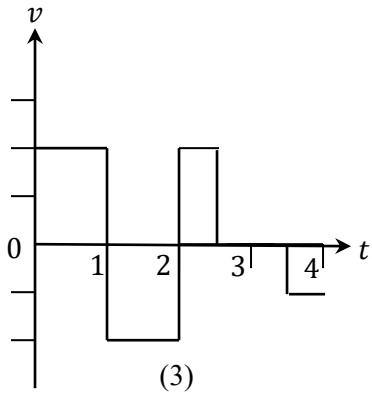




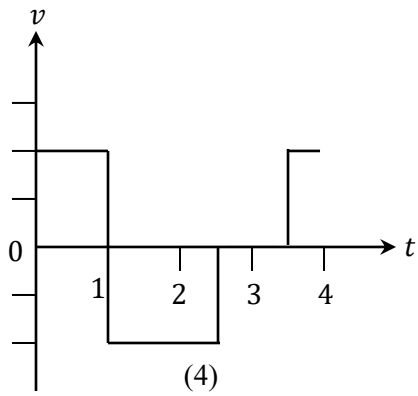
(1)



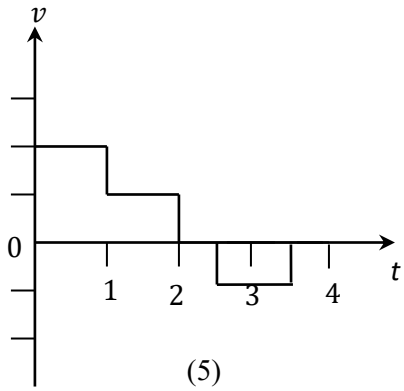
(2)



(3)



(4)

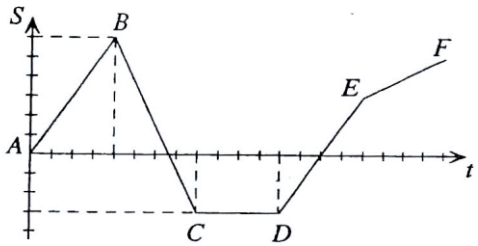


(5)

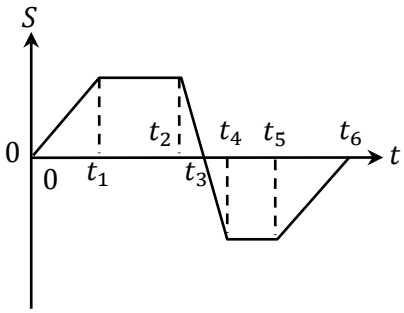
12. Displacement (S) of a particle measured along the x – axis with time (t) is shown in the figure. The magnitude of the velocity of the particle is largest when it travels from

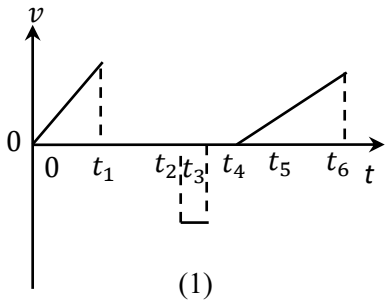
(1) A to B (2) B to C (3) C to D

(4) D to E (5) E to F

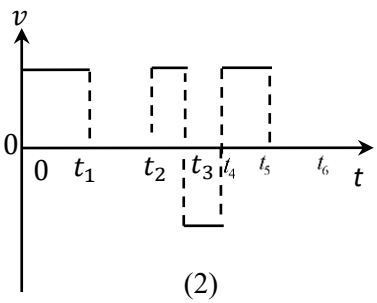


13. The displacement (S) – time (t) curve of an object is shown in the figure. The corresponding velocity (v) – time (t) curve is best represented by

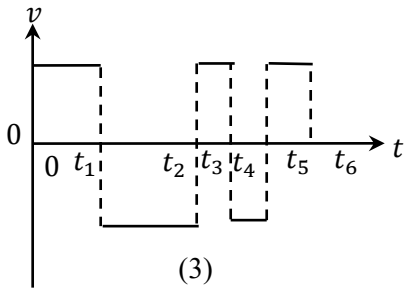




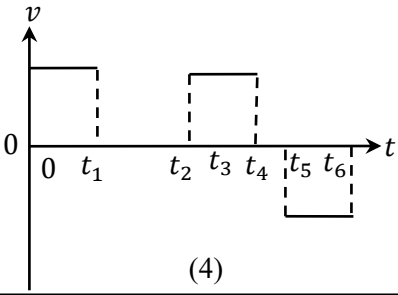
(1)



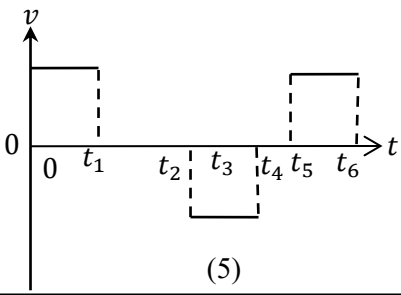
(2)



(3)

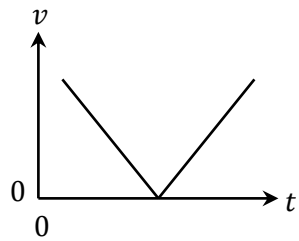


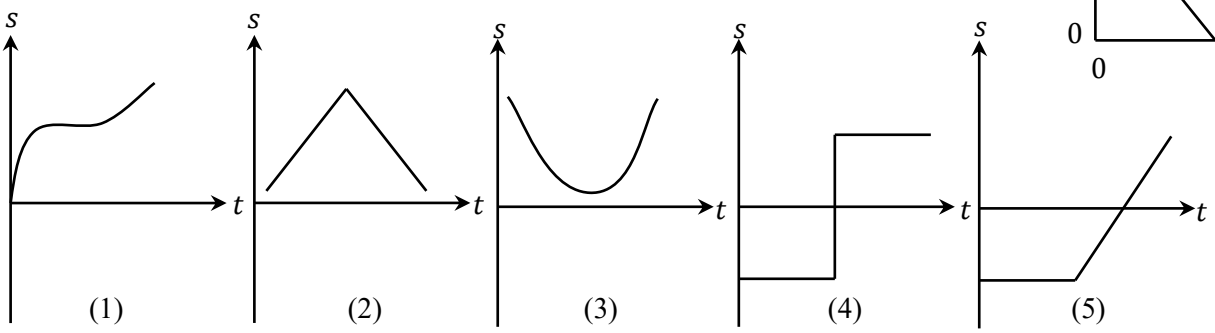
(4)



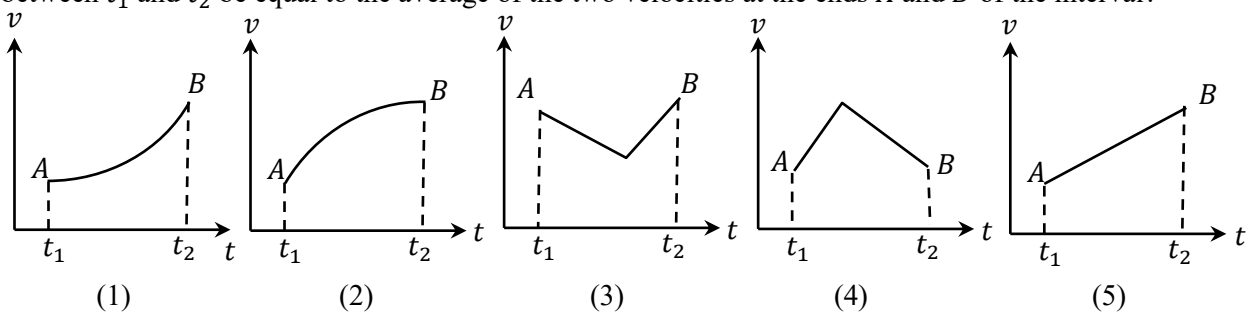
(5)

14. The graph shown represents the velocity (v) – time (t) curve for an object. The corresponding displacement (s) – time (t) curve is best represented by

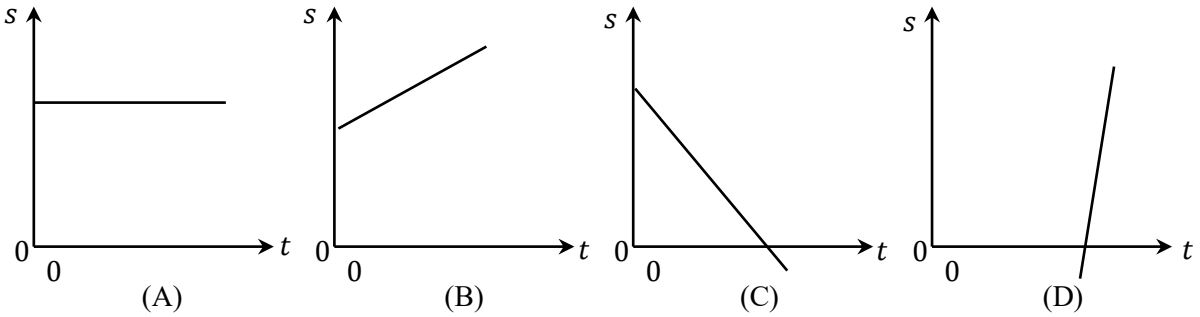




15. In which of velocity (v) – time (t) graphs shown below would the average velocity over the entire period between t_1 and t_2 be equal to the average of the two velocities at the ends A and B of the interval?



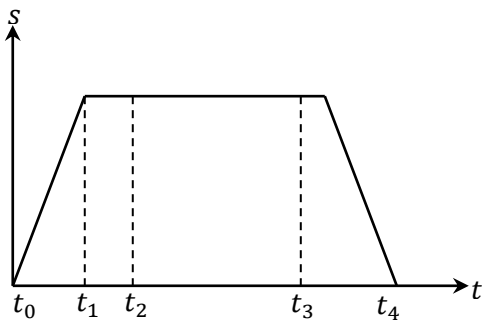
16.



Of the given distance (s) – time (t) graphs drawn to same scale, the magnitude of the velocity is

(1) minimum in A and maximum in C. (2) minimum in C and maximum in D.
(3) minimum in A and maximum in D. (4) minimum in B and maximum in C.
(5) minimum in D and maximum in B.

17. Figure shows displacement (s) versus time (t) curve for a motion of a particle. Consider the following statements made about its motion.

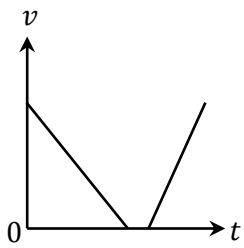


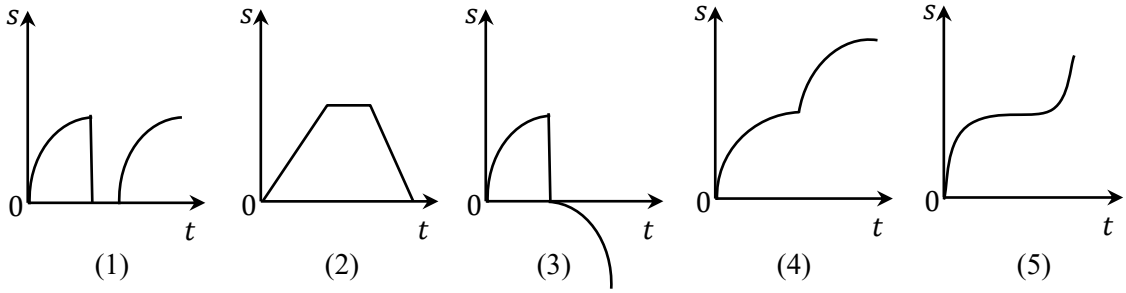
(A) During the time period $t_0 - t_1$ the particle moves at a constant acceleration and during $t_2 - t_3$, it moves at a constant velocity.
(B) Particle comes to rest at time t_4.
(C) During the time period $t_0 - t_4$, the total distance traveled by the particle is equal to the area under the $s - t$ curve.

Of the above statements,

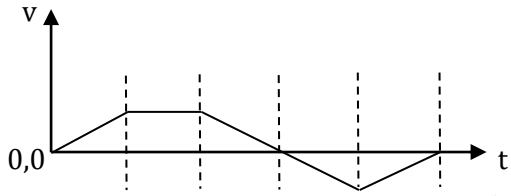
(1) only (A) is true. (2) only (A) and (B) are true. (3) only (B) and (C) are true.
(4) all (A), (B) and (C) are true. (5) all (A), (B) and (C) are false.

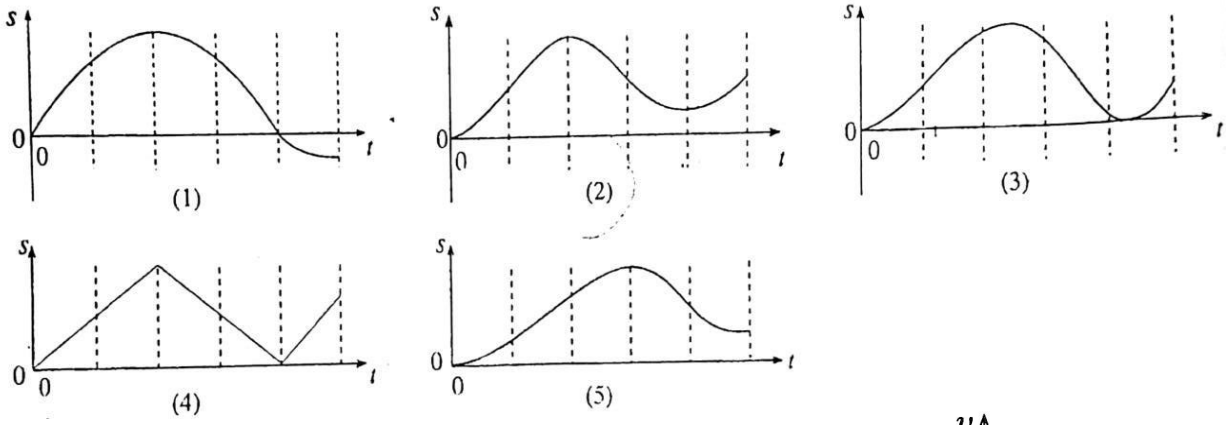
18. Velocity (v) - time (t) curve for the motion of a particle is shown in the figure. The corresponding displacement (s) - time (t) curve would be





19. The variation of the velocity (v) with time (t) of a particle is shown in the figure. The corresponding displacement (s) S – time (t) curve is best represented by





20. If the velocity (v) of an object varies with time (t) as shown in figure (a), the corresponding variation of the displacement (x) with time (t) is best represented by

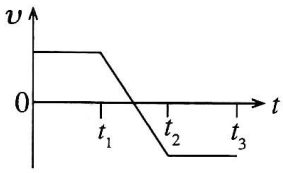
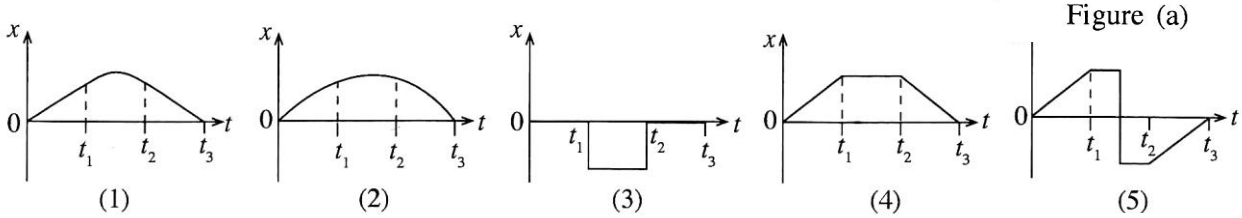
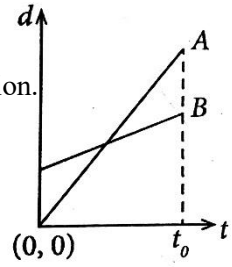


Figure (a)

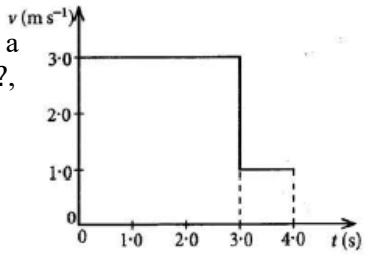


21. Two straight lines shown in the displacement $(d) -$ time (t) graph represent the motions of two objects A and B started from rest at time $t = 0$ and moving along the positive $x -$ direction. Which of the following statements made about the motions of the objects is true?



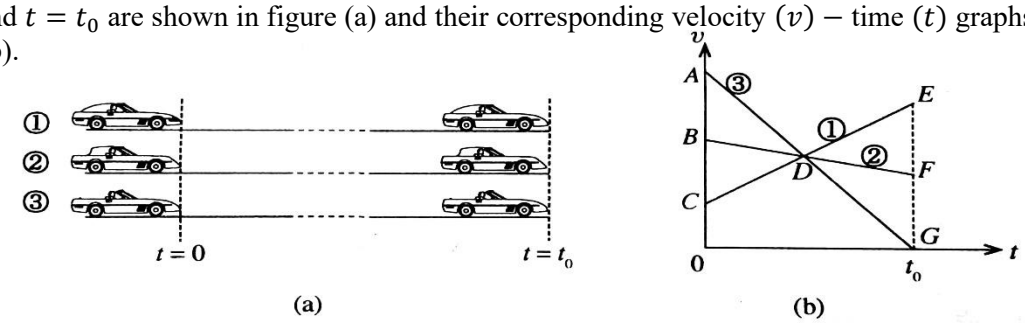
(1) The object A has travelled for a longer time than B.
(2) When $t = t_0$ object B has made a displacement greater than A.
(3) Object A has a greater velocity than B.
(4) Object A has a greater acceleration than B.
(5) Both objects have the same velocity at the point where the two straight lines cross each other.

22. The figure shows the velocity (v)- time(t) graph for an object moving along a straight line. What is the average velocity of the object from $t = 0$ to $t = 4$?,



(1) $1.5\ \mathrm{ms^{-1}}$ (2) $2.0\ \mathrm{ms^{-1}}$
(3) $2.5\ \mathrm{ms^{-1}}$ (4) $2.7\ \mathrm{ms^{-1}}$
(5) $3.3\ \mathrm{ms^{-1}}$

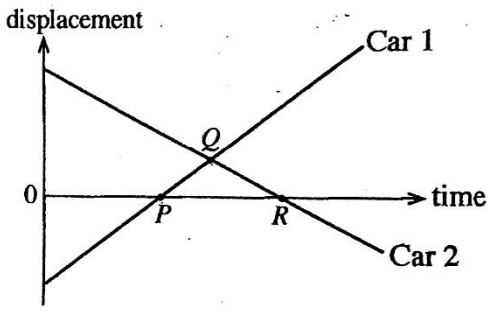
23. Positions of three motor cars ①, ② and ③ travelling along three parallel straight lanes of a road at time $t = 0$ and $t = t_0$ are shown in figure (a) and their corresponding velocity (v) – time (t) graphs are shown in figure (b).



The situation shown in figure (a) could have happened only if the areas in the graphs satisfy the conditions

(1) $ABD = DEF$ and $ABD = DEG$
(2) $BCD = DEF$ and $ABD = DFG$
(3) $CDB = DEG$ and $ABD = DEF$
(4) $BCD = ABD$ and $DEF = DFG$
(5) $CD = DFG$ and $BCD = DFG$

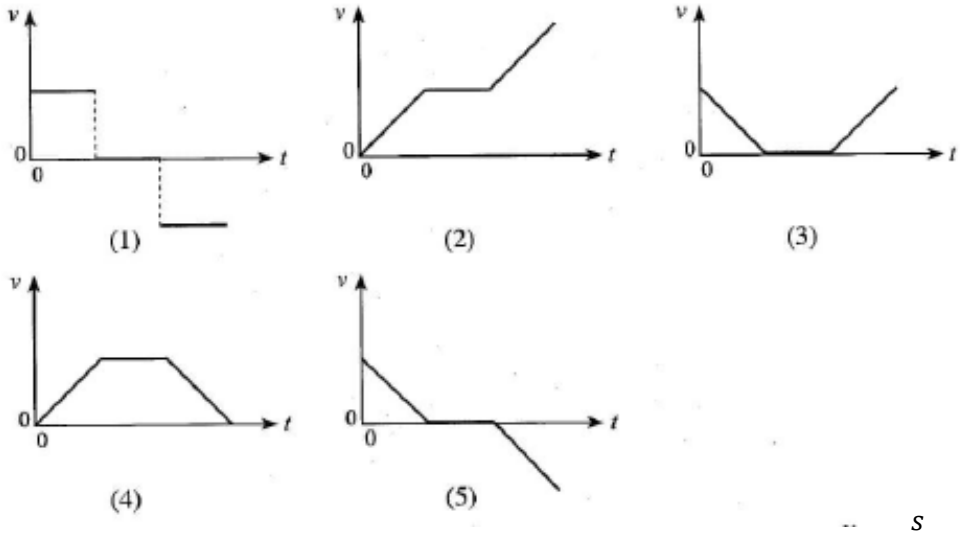
24. The figure shows the displacement – time graphs for the motion of two cars with respect to a lamp post aside the road. Consider the displacement to the right side of the lamp post as positive. A student has made the following statements regarding the motion of cars relevant to the points P, Q and R marked on the graph.



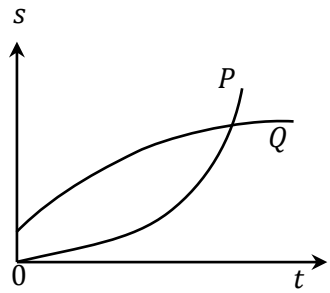
(A) Relevant to P: Car 1 coming from left crosses Car 2.
(B) Relevant to Q: Both cars are moving towards the lamp post and cross each other.
(C) Relevant to R: Car 2 moving from right passes the lamp post.
Which of the above statements is/are correct?

(1) Only (B). (2) Only (C). (3) Only (A) and (B).
(4) Only (B) and (C). (5) All (A), (B) and (C).

25. The displacement-time $(s - t)$ graph for the motion of an object is shown in the figure. The corresponding velocity – time $(v - t)$ graph would be best represented by,



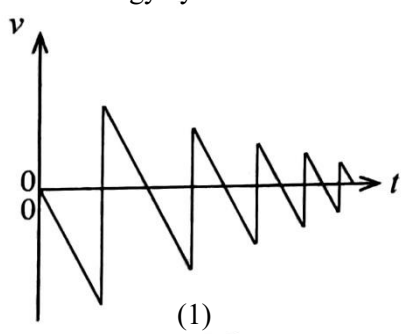
26. The figure shows the displacement (s) – time (t) graphs of two objects P and Q moving along a straight line. Consider the following statements.



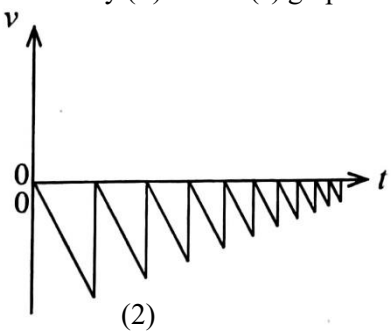
(A) Both objects have velocities in the same direction.
(B) Velocities of both objects increase with time.
(C) Both objects have the same velocity at the crossing point of the two graphs.
Of the above statements,

(1) only (A) is true. (2) only (A) and (B) are true.
(3) only (A) and (C) are true (4) only (B) and (C) are true.
(5) all (A), (B) and (C) are true.

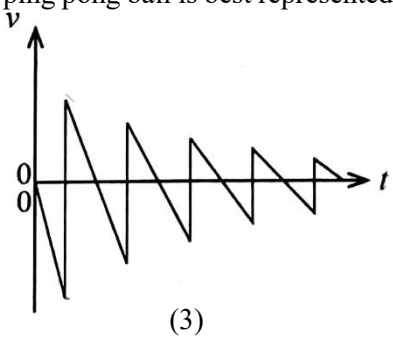
27. A ping pong ball is dropped from a height of 1 m above a table. In every successive bounce, the ball loses its kinetic energy by the same amount. The velocity (v) – time (t) graph of the ping pong ball is best represented by,



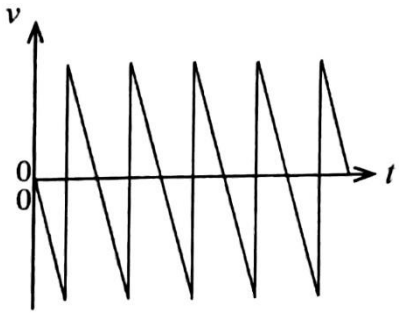
(1)



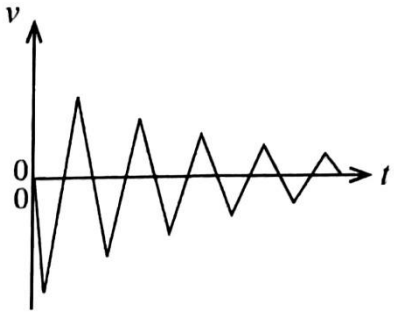
(2)



(3)



(4)



(5)

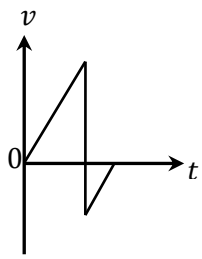
28. A ball is thrown straight up from the edge of the roof of a building, which is 20 m above the ground and upon coming down the ball just misses the edge of the roof and reaches the ground. If the entire journey takes 4 s, the initial velocity with which the ball was thrown up is,
(1) $1\ \mathrm{m\ s^{-1}}$ (2) $5\ \mathrm{m\ s^{-1}}$ (3) $10\ \mathrm{m\ s^{-1}}$ (4) $15\ \mathrm{m\ s^{-1}}$ (5) $20\ \mathrm{m\ s^{-1}}$

29. An object is projected upwards with a velocity of $100\ \mathrm{m\ s^{-1}}$ from the ground. If the air resistance is neglected it will strike the ground in,
(1) 5 s (2) 10 s (3) 1.5 s (4) 20 s (5) 25 s

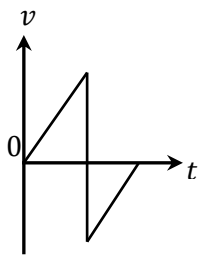
30. An object is dropped from a helicopter which is moving horizontally at a constant velocity of $45\ \mathrm{m\ s^{-1}}$, 180 m above the ground. Time taken for the object to reach the ground is
(1) 3 s (2) 4 s (3) 5 s (4) 6 s (5) 12 s

31. A ball thrown vertically upwards returns to the thrower's hand. Consider the following statements
(A) If there is no air resistance, the times taken for the upward and the downward journeys are the same.
(B) If there is air resistance, the ball will reach the thrower's hand with a speed that is less than the speed with which it was thrown up.
(C) If there is air resistance, the time taken for the upward journey is greater than that for the downward journey.
Of the above statements,
(1) only (A) is true. (2) only (B) is true (3) only (C) is true.
(4) only (A) and (B) are true. (5) all (A), (B) and (C) are true.

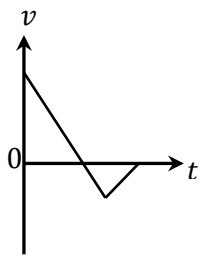
32. An object dropped from a height h bounces back from the floor to a height $\frac{h}{2}$. Which of the following velocity (v) – time (t) graphs best represents the motion of the object?



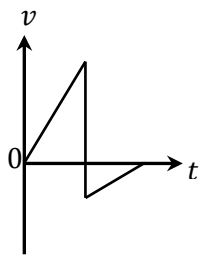
(1)



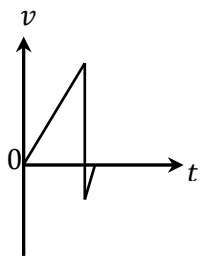
(2)



(3)



(4)

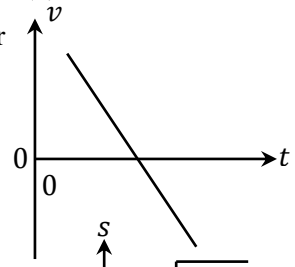


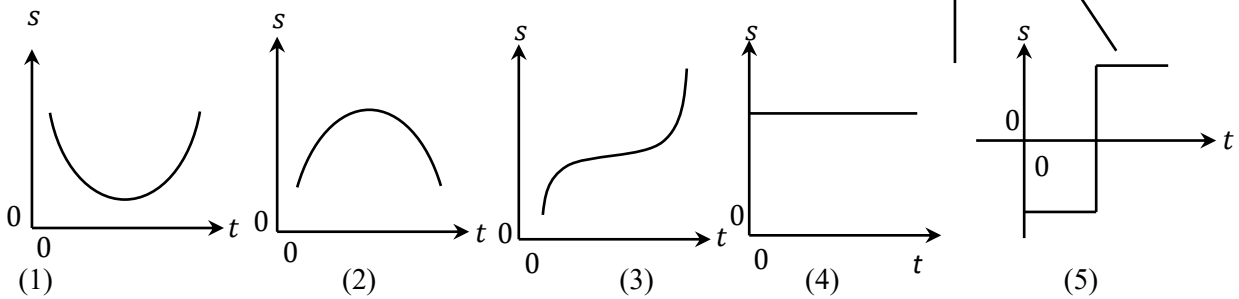
(5)

33. A bird flying at a height of 40 m with a speed of $10\ \mathrm{m\ s^{-1}}$ drops a small fruit from its mouth. If free fall is assumed the speed of the fruit just before it reaches the ground is
(1) $10\ \mathrm{m\ s^{-1}}$ (2) $15\ \mathrm{m\ s^{-1}}$ (3) $20\sqrt{2}\ \mathrm{m\ s^{-1}}$ (4) $25\ \mathrm{m\ s^{-1}}$ (5) $30\ \mathrm{m\ s^{-1}}$

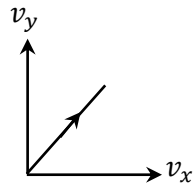
34. The distances travelled by an object falling freely from rest during first, second and third seconds are in the Ratio
(1) $1:2:3$ (2) $1:4:9$ (3) $1:2:9$ (4) $1:1:1$ (5) $1:3:5$

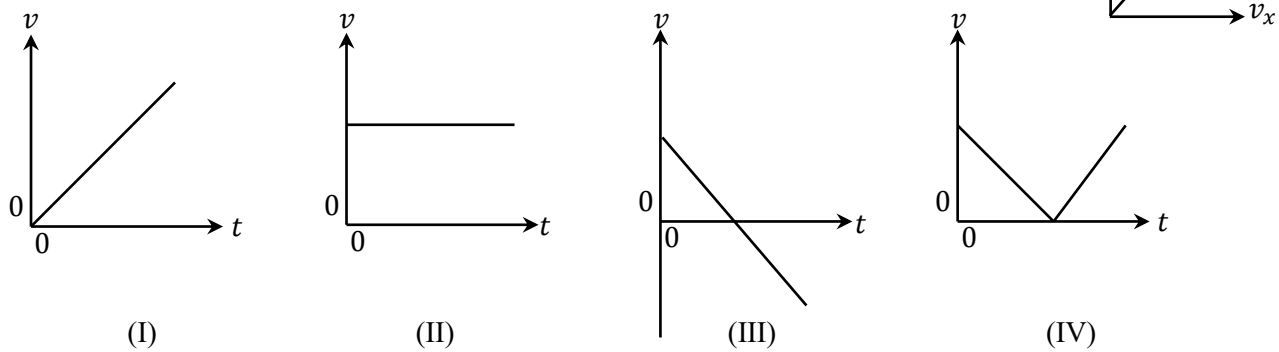
35. The graph shown in the figure represents the velocity (v) – time (t) curve for an object. The corresponding displacement (s) – time (t) curve is best represented by





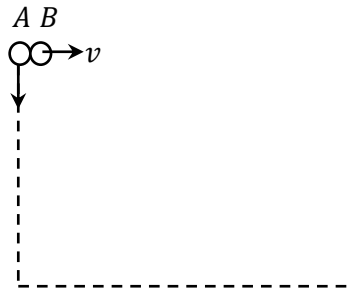
36. A stone is thrown at a certain angle with the horizontal in the direction shown by the arrow. If air resistance is ignored, which of the following velocity (v) – time (t) graphs best represent the variations of v_x with t, and v_y with t?





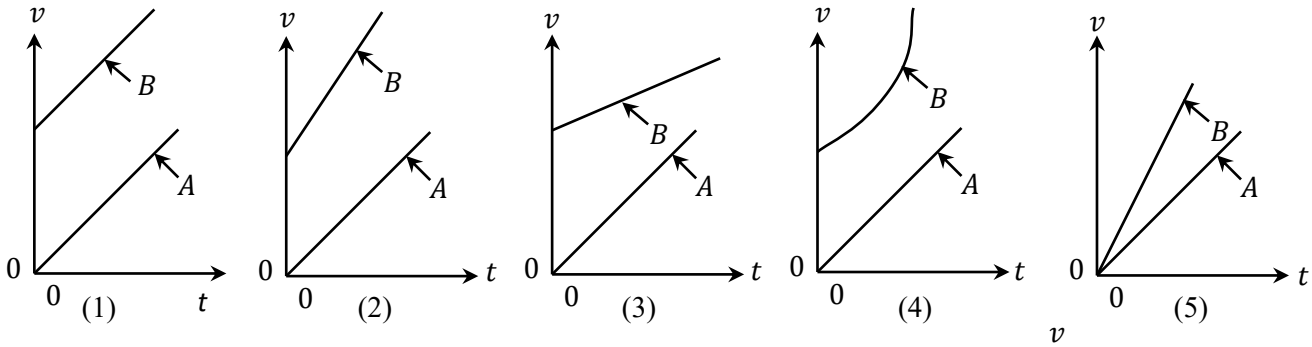
	v_x with t	v_y with t
(1)	II	III
(2)	II	I
(3)	I	IV
(4)	II	IV
(5)	II	II

37. A ball B is projected horizontally with speed v and a ball A is dropped vertically from rest at the same instant as shown in the figure. Which of the following statements is true? (Neglect air resistance)

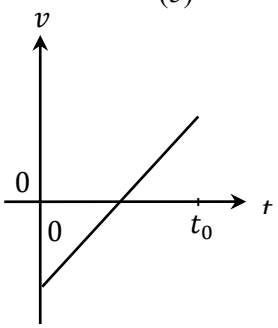


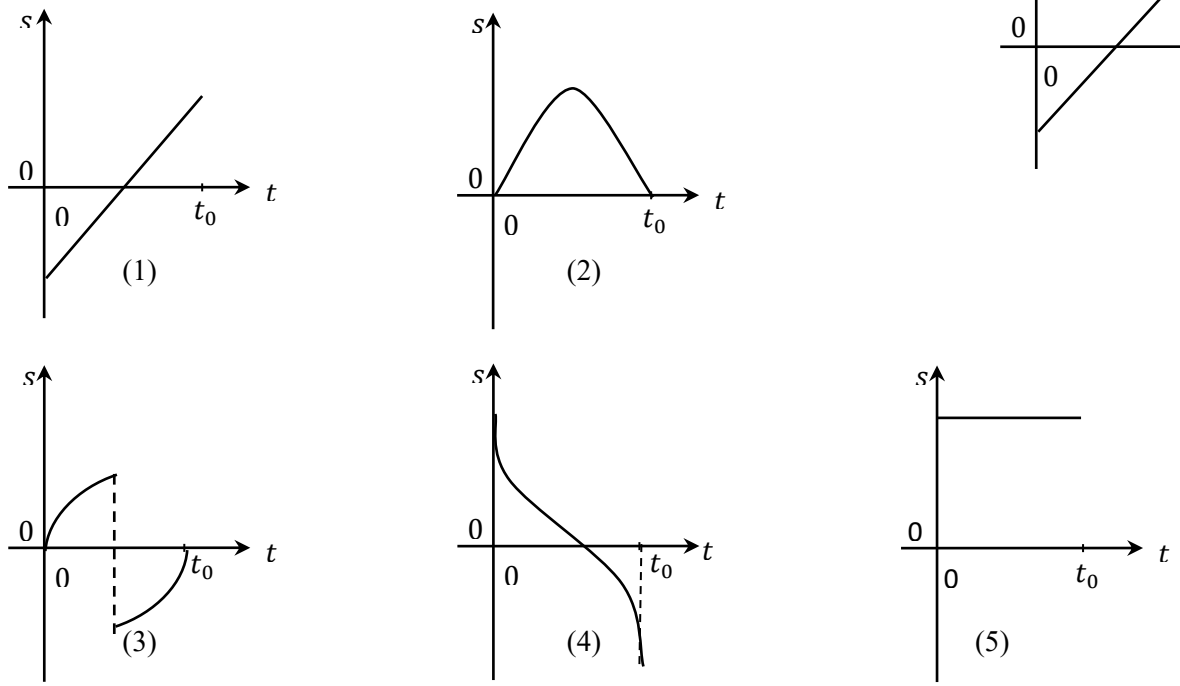
(1) A reaches the ground first with a higher speed than B.
(2) B reaches the ground first with a higher speed than A.
(3) A reaches the ground first with a lower speed than B.
(4) Both A and B reach the ground at the same instant with the same speed.
(5) Both A and B reach the ground at the same instant but B with higher speed than A.

38. A person simultaneously drops an object and throws another object vertically downwards from a certain height. Which of the following graphs best represents the velocity (v) – time (t) curves for the two objects? (Curve A represents the dropped object and curve B represents the thrown object)



39. Variation of the velocity v of an object with time t is shown in the figure. The corresponding variation of the displacement s of the object with time t is best represented by



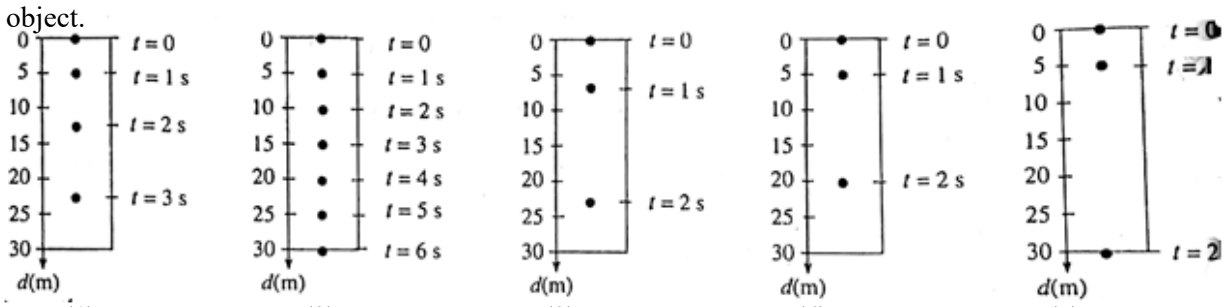


40. A particle is projected vertically up along the Y axis from the origin ($y = 0$) at $t = 0$ with a velocity v_1, and at the same instant another particle is projected vertically downwards along the Y axis from a height H ($y = H$) with a velocity v_2. If the two particles meet each other at a height $y = \frac{H}{2}$ at time $t = t$ then

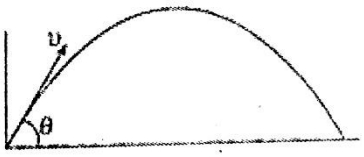
(1) $v_1^2 - v_2^2 = g^2t^2$ (2) $v_1^2 - v_2^2 = 2gH$ (3) $v_1 - v_2 = \frac{1}{2}gt$

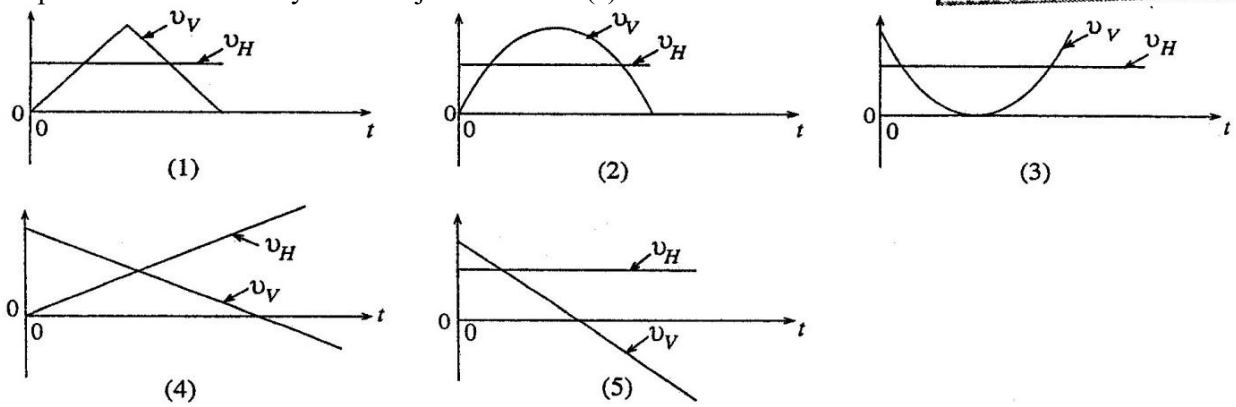
(4) $v_1 - v_2 = 2gH$ (5) $v_1 - v_2 = gt$

41. Photographs of an object that starts falling freely from rest at $t = 0$ are taken by a camera, first at $t = 0$ and thereafter at the end of each second. Which of the following diagrams correctly indicates the location of the object at the end of each second? The vertical axes of the diagrams represent the distance (d) travelled by the object.



42. An object is projected under gravity with velocity v, in a direction which makes an angle θ with the horizontal as shown in the figure. Which of the following graphs correctly indicates the variation of the horizontal (v_H) and vertical (v_V) components of the velocity of the object with time (t)?





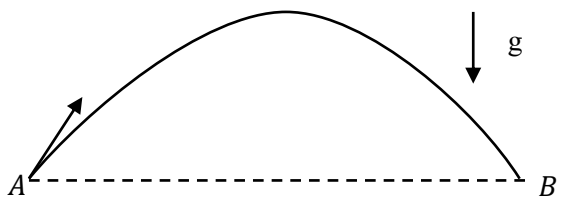
43. A ball is dropped freely to a floor from a height of 1 m. If its speed is reduced by 25 % at each bounce, what would be the height the ball reaches after three bounces?

(1) $\frac{3}{4}$ m (2) $\left(\frac{3}{4}\right)^2$ m (3) $\left(\frac{3}{4}\right)^3$ m (4) $\left(\frac{3}{4}\right)^6$ m (5) $\left(\frac{3}{4}\right)^9$

44. A ball is thrown vertically upward from the ground. The time duration between the two instances that the ball passes the point at a height of 25m on its path from the ground is 4s. What is the initial velocity of the ball? (Neglect air resistance)

(1) 20 ms^{-1} (2) 25 ms^{-1} (3) 30 ms^{-1} (4) 35 ms^{-1} (5) 40 ms^{-1}

45. The figure shows the path AB of a ball projected from point A. Neglect the air resistance. Consider the following statements.



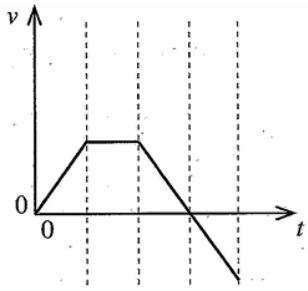
(A) The velocity of the ball at the maximum height of the path is zero.
(B) The velocity of the ball at point B is equal to the velocity at point A.
(C) The kinetic energy of the ball at point B is equal to the kinetic energy at point A.
Of the above statements,

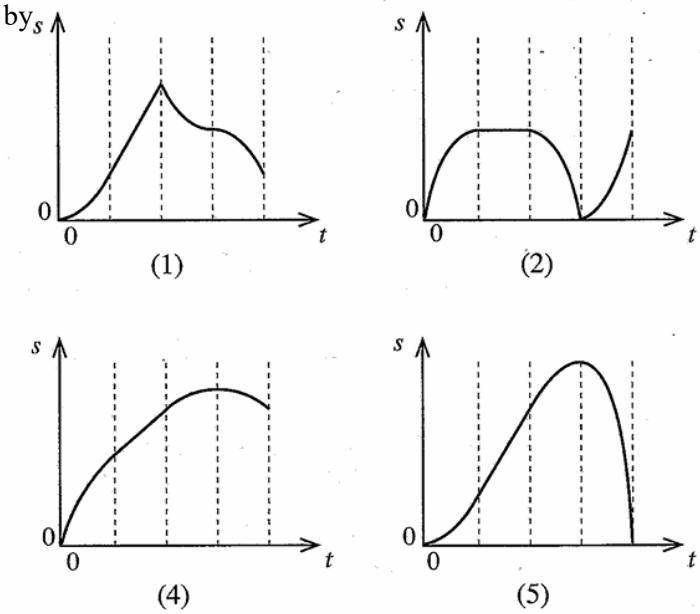
(1) Only (A) is true.
(3) Only (C) is true.
(4) Only (B) and (C) are true.
(4) Only (A) and (C) are true.
(5) All (A), (B) and (C) are true.

46. An object is projected with an initial velocity u at an angle θ to the horizontal. Neglect air resistance. Consider the following statements regarding the projectile.
(A) The kinetic energy of the object remains constant throughout the motion.
(B) The velocity of the object is zero at the highest point of the trajectory.
(C) The acceleration of the object is constant throughout the motion.
Of the above statements,

(1) Only (A) is true.
(3) Only (C) is true.
(4) Only (B) and (C) are true.
(4) Only (A) and (C) are true.
(5) All (A), (B) and (C) are true.

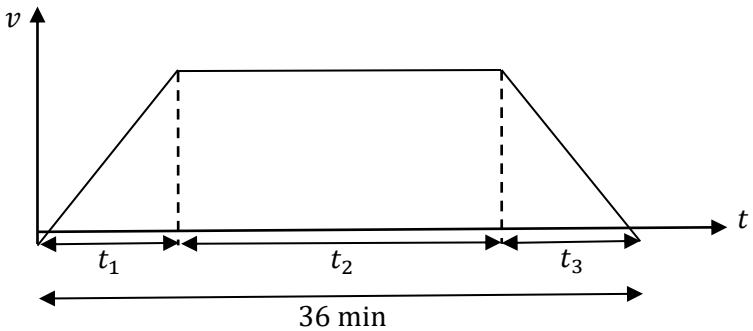
47. The graph of the variation of velocity (v) with time (t) of an object is shown in the figure. The corresponding displacement (s) -time (t) curve is best represented by





48. A body is dropped freely from a height above the ground. At the same time another body is thrown horizontally from the same height. Assume that air resistance is negligible. Which of the following statements is true regarding the time of flight of the two bodies?
(1) Cannot answer without the value of the initial velocity of the thrown body.
(2) Cannot answer without the value of the height
(3) The thrown body will take a longer time to hit the ground.
(4) The thrown body will hit the ground first
(5) Both bodies will hit the ground at the same time

49. The velocity (v) -time (t) graph for the motion of a train between two railway stations x and y is shown in the figure. The distances travelled by train during time intervals t_1, t_2 and t_3 are 3 km, 11 km and 2 km respectively. The maximum velocity of the train in kmh^{-1} is
(1) 27
(2) 30
(3) 35
(4) 40
(5) 42



50. The velocity of a car traveling at a uniform rate of deceleration decreases from 30ms to 15ms in 10s. After how much time does it come to rest?
(1) 5s (2) 10s (3) 12s (4) 15s (5) 20s

Error Analysis

Question No.	1st attempt					2nd attempt					3rd attempt				
	E1	*E2*	*E3*	*E4*	*E5*	*E1*	*E2*	*E3*	*E4*	*E5*	*E1*	*E2*	*E3*	*E4*	*E5*
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